

AI in Genomics 2026

Tunisia · 14–18 Sept 2026

The Transformative Role of AI in the Post-Genomic Era

Presenter: Jame Tannous

Andrés Posbeyikian, Yu Sugihara, Amal Boukteb,
Jame Tannous, James Canham, Sophien Kamoun



getgenome@tsl.ac.uk · getgenome.net · *#AIGenomics2026*

The data deluge in genomics

More data. Greater insights.
Bigger challenges.

How do we
store, analyse
and make sense
of all this?



Millions
of genomes



Terabases
of sequence data



Multi-omics
datasets

Genomics
Transcriptomics
Proteomics
Epigenomics
Metabolomics
Single-cell
Spatial



Growing
public databases

SRA
ENA
NCBI
...



Global
collaboration

More researchers
More data
More opportunities



From gigabytes
to petabytes



From individual projects
to global consortia



From single data types
to multi-omics



A data deluge today.
A healthier tomorrow.

Learning Outcomes

- **Understand genome annotation: Input, output, and workflows**
- **AI models: Types, advances, and limitations**

BASICS OF BIOINFORMATICS

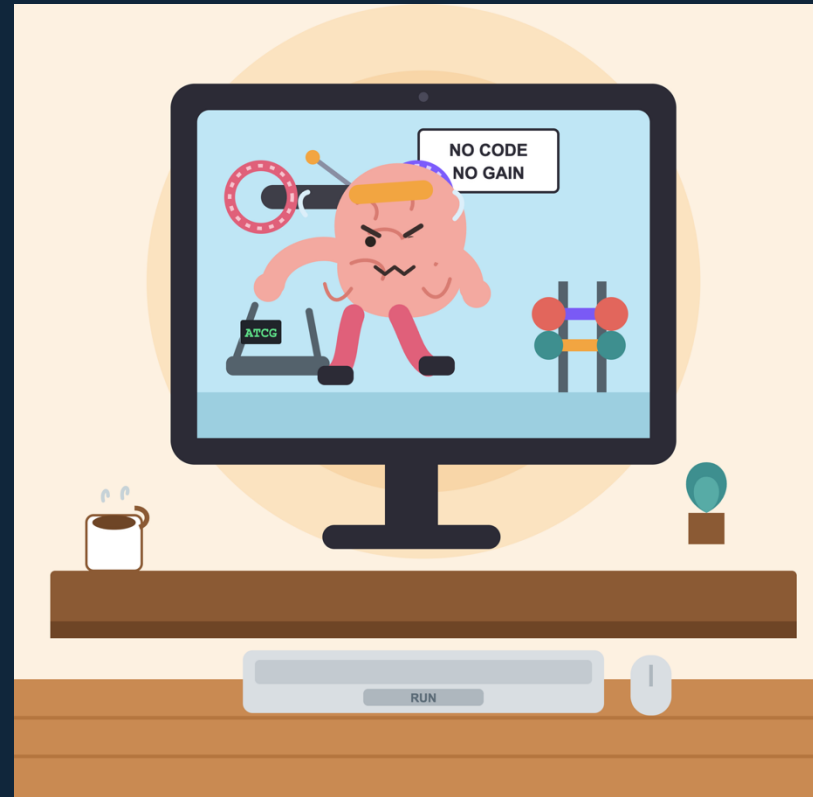
Master the Command-line

Python, R, UNIX, and more

<https://astrobiomike.github.io/>

> Tunis R User Group

Terminal – Directory – Arguments – Files – Input – Output



TUNIS R USER GROUP



Members Projects About Webinars R/Medicine News Code of Conduct

Bridging Gaps: Tunis R User Group's Journey in Democratizing R in Bioinformatics

By R Consortium | April 26, 2024 | Blog



Last year, the R Consortium had a conversation with Amal Tlili, the co-organizer of the [Tunis R User Group](#), regarding the [Use of R for Marketing and CRM in Tunisia](#). This year, [Amal Boukteb](#) and [Hedia Tnani](#) spoke to the R Consortium about the use of R for bioinformatics research in Tunisia and discussed the group's efforts to bridge the gap between academia and industry. The Tunis R User Group hosts engaging virtual events to connect R enthusiasts across the MENA region and Worldwide. Their events promote the use of R and foster knowledge and skill development in data science and bioinformatics.



Amal Boukteb
Aspiring Bioinformatician



Hedia Tnani
Bioinformatician/Data Analyst



tunis.rusergroup.org/
Twitter LinkedIn



Dr Amani Ben Alaya
Centre of Biotechnology
of Borj Cedria



Dr Manel Chaouachi
Centre of Biotechnology
of Borj Cedria



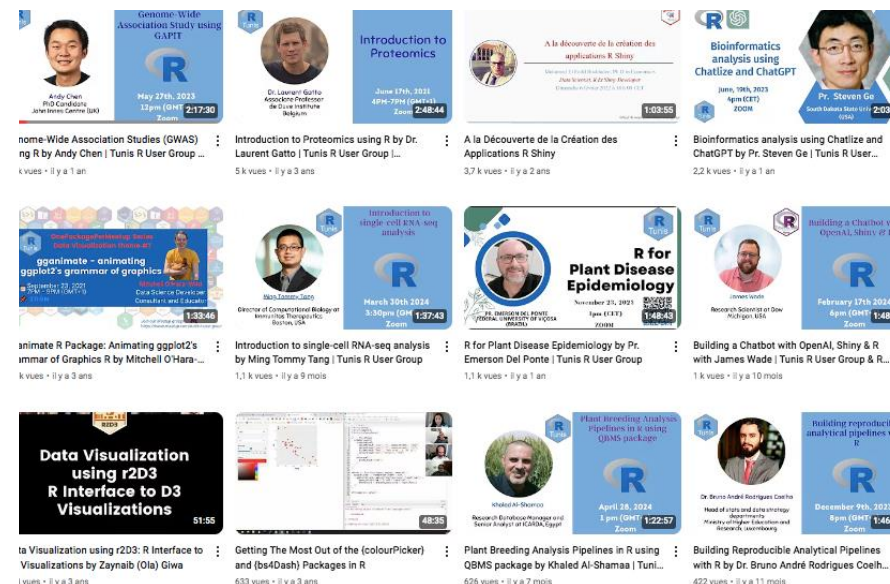
Tunis R User Group

@TunisRUserGroup • 1,63 k abonnés • 16 vidéos

Tunis R User Group is part of an international mentorship group with a focus on helping m

meetup.com/tunis-r-user-group et 5 autres liens

Abonné



TUNIS R USER GROUP



Bioinformatics analysis
using ChatLize and ChatGPT

June, 19th, 2023 4pm (CET)

ZOOM



600 registrations



Pr. Steven Ge

South Dakota State University
(USA)



Andy Chen
PhD Candidate
John Innes Centre (UK)

Genome-Wide
Association Study using
GAPIT



May 27th, 2023

12pm (GMT)

Zoom

2:17:30

Genome-Wide Association Studies (GWAS) :
using R by Andy Chen | Tunis R User Group ...

18K views



@TunisRUserGroup



@TunisRUserGroup



tunis.rusergroup@gmail.com



Tunis R User Group

★★★★★ 4.8 • 57 ratings

📍 Tunis, Tunisia

👤 1,038 members

TUNIS R USER GROUP



THE
CARPENTRIES

Bioconductor Carpentries Workshop

Introduction to genomic data analysis with R and Bioconductor

National Institute for Agricultural Research of Tunisia

Oct 30-31, 2024

9:00 - 16:30 CET

Instructors: Amal Boukteb, Hédia Trani,
Malek Chaouch

Helpers: Manel Chaouachi, Ameni Ben
Alaya



BIOINFORMATICS RESOURCES · DEEP DIVE

Artificial Intelligence and Machine Learning

Understanding ML models and their applications in genomics

Genomes · Genes · Proteins

Five Core AI Approaches Used in Genomics

Different genomic questions call for different types of AI — here's how the five major categories map to real tasks.

1

Machine Learning

Makes predictions based on patterns in available data.

e.g. Distinguishing disease-causing variants from benign ones

2

Deep Learning

Multi-layer neural networks for complex pattern recognition.

e.g. Analyzing images for diagnosis; identifying mutations; predicting variant impact; designing guide RNAs (gRNAs) for gene editing (**AlphaFold**)

3

Natural Language Processing (NLP)

The human–computer interface; understands and interprets text.

e.g. Analyzing electronic health records and antibiograms (ChatGPT is an NLP system)

4

Generative Adversarial Networks (GANs)

Two competing networks that generate and/or interpret new data.

e.g. Art & image generation, image-to-image translation, data augmentation, designing novel molecules for drug discovery

5

Reinforcement Learning

Learns optimal decisions via trial, error, and reward.

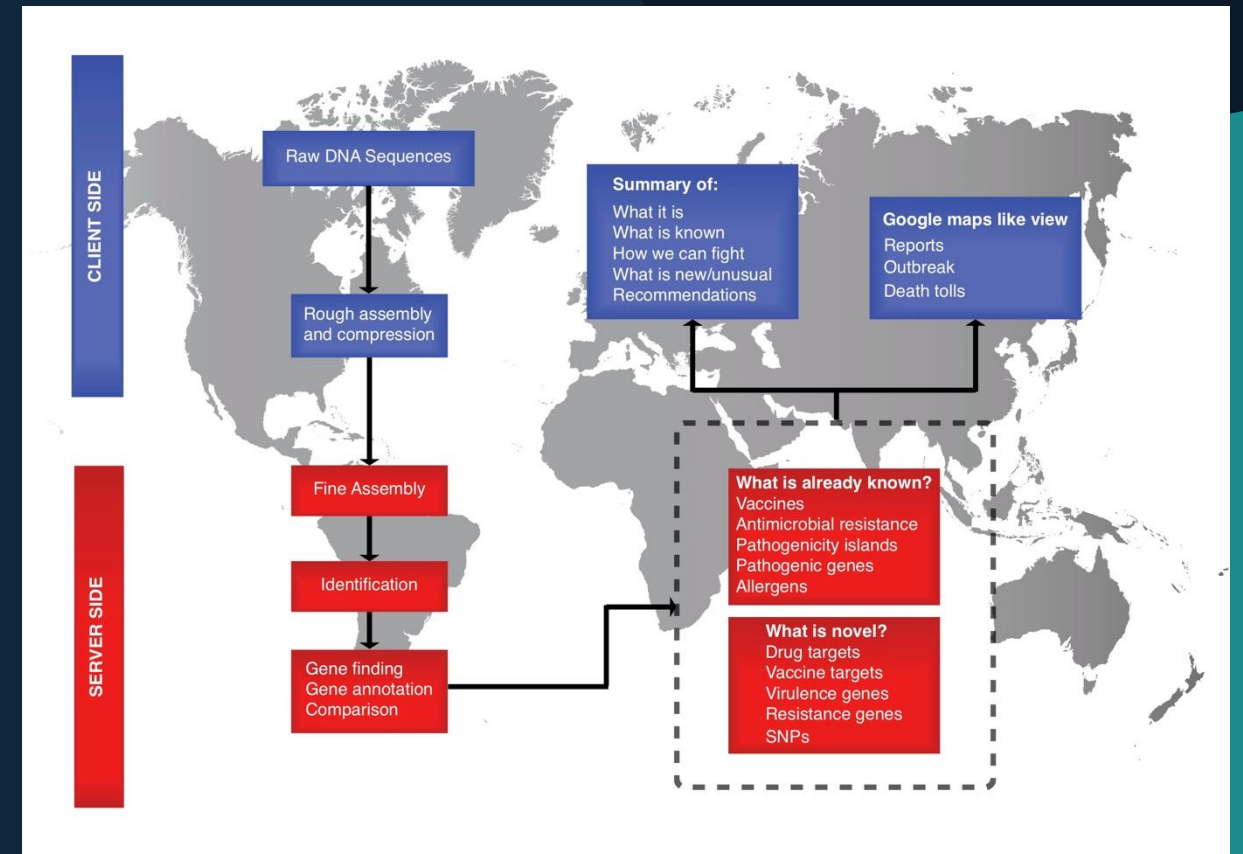
e.g. Optimizing DNA/protein sequences, personalized treatment strategies, inferring gene regulatory networks, improving experimental protocols

AN ALL-IN-ONE PLATFORM

The Center for Genomic Epidemiology

Genome mining in one click

From nucleotide bases to meaningful information



A COMPREHENSIVE DATABASE

Genome Annotation

Making sense of the ATCG file



From nucleotide bases to meaningful information

Prokka vs. RAST: A Comparison

Prokka	 Get Genome GENOMICS FOR ALL	RAST
CLI & GUI		GUI
Loop for batch annotation		Manual, one-by-one, upload
Annotation time depends on genome size		Annotation time depends on genome size and job load
Output files (gbk, gff, fna, tsv, tbl, sqn...)		Web visualization
Downstream analyses (phylogenetic tree, pangenome, visualizations)		Clustering into subsystems (membrane transport, DNA replication and repair, transcriptional regulation)

Prokka's Output Files, at a Glance

.gff

Master annotation (GFF3): sequences + features together

.gbk

Standard GenBank flat file, derived from the .gff

.fna

Nucleotide FASTA of the input contigs

.faa

Protein FASTA of translated CDS sequences

.ffn

Nucleotide FASTA of all annotated features (not just CDS)

.tsv

Tab-separated summary table of every annotated feature

.txt

Plain-text statistics: counts of each feature type

.sqn/ .fsa/ .tbl

GenBank/Sequin submission files (via tbl2asn)

.err

NCBI discrepancy report of unacceptable annotations

.log

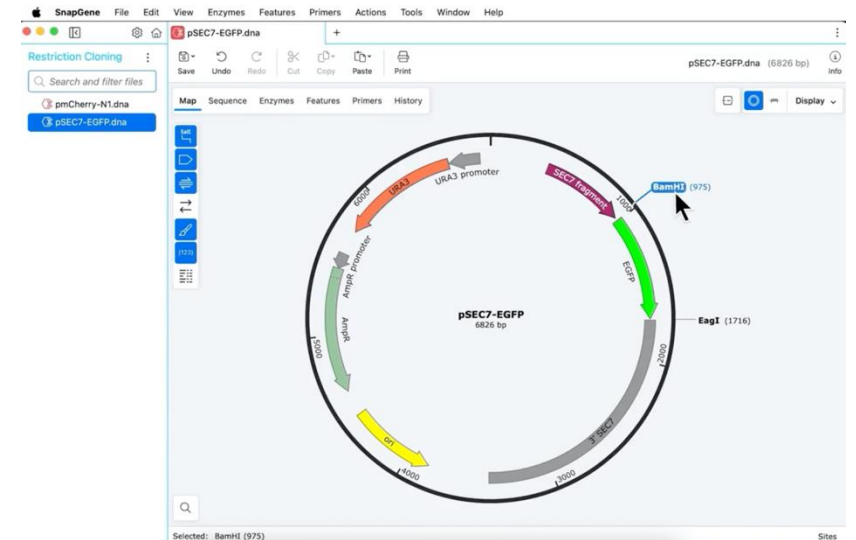
Full run log — settings used for this Prokka job

GBK & GFF: The Two Workhorse Formats

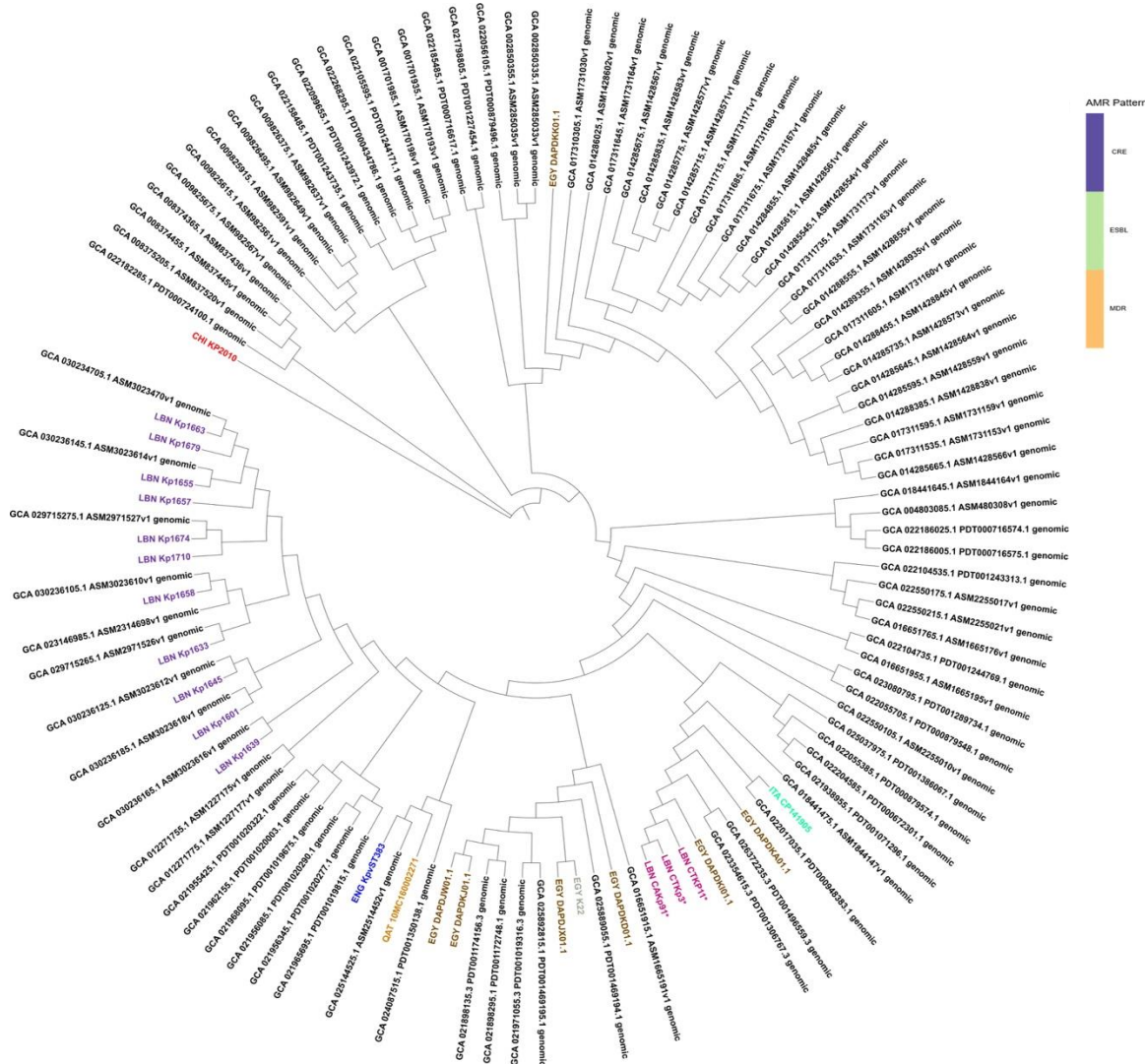


.gbk — Visualization & Genomic Context

- Rich flat file: sequence + annotations + metadata in one record
- Viewable directly in SnapGene, Artemis, or IGV
- Preferred input for tools like clinker, which builds publication-ready gene-cluster comparison figures showing the genetic environment around genes of interest



GBK & GFF: The Two Workhorse Formats



.gff — Structural Data & Pangenome Construction

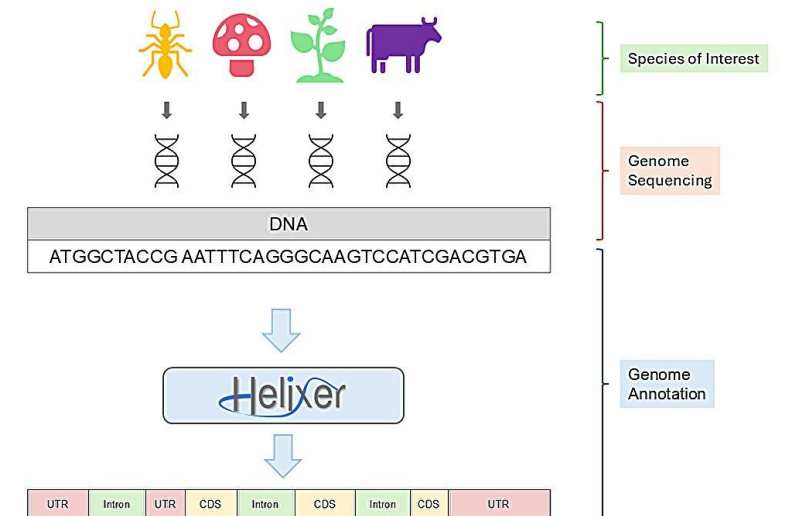
- Lightweight, tab-delimited structural feature format
- The standard input for pangenome pipelines: Roary and Panaroo both take Prokka's GFF3 output directly
- Enables fast, large-scale comparison of gene content across hundreds–thousands of genomes

Helixer



- The final GFF3 contains primary gene models with gene, mRNA, exon, CDS, and 5'/3' UTR — ready to view in a genome browser or feed into downstream tools (e.g. gffread to derive protein/transcript FASTA).
- Helixer predicts one transcript per gene — it does not model alternative splicing isoforms and mRNA feature counts are always equal.

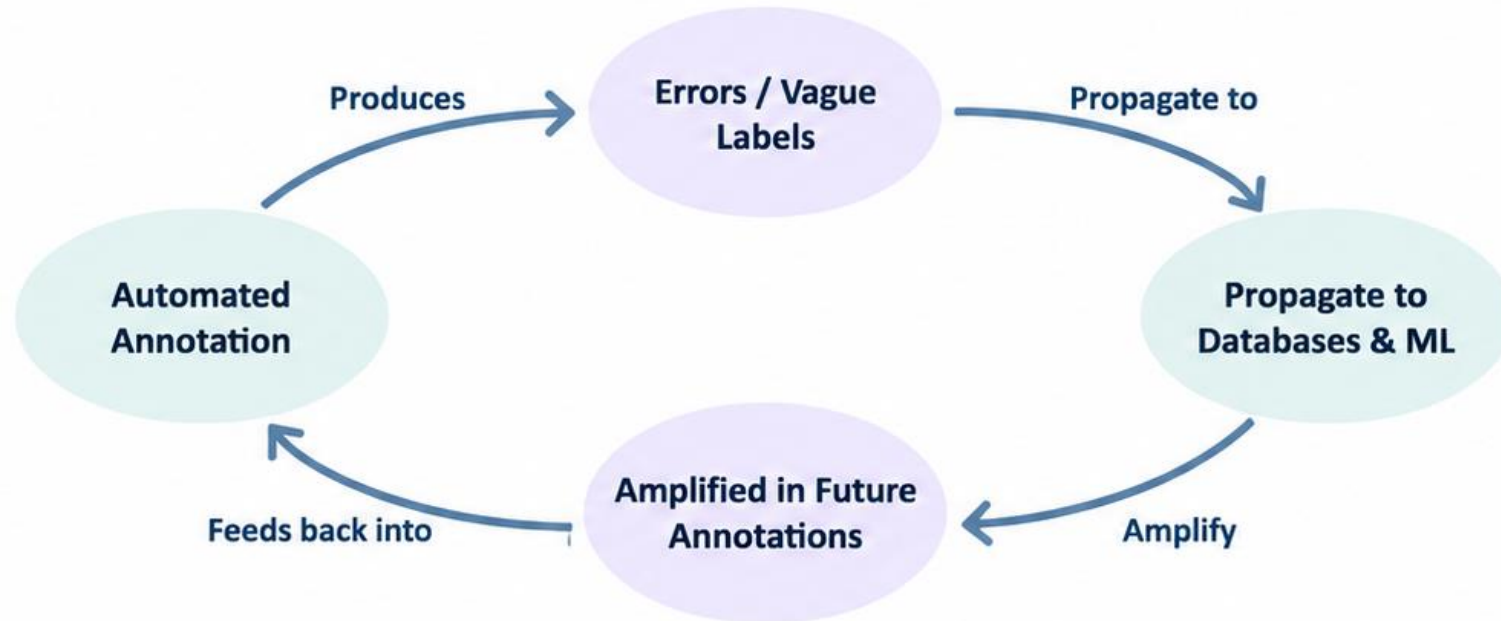
e.g. `grep -v "^#" output.gff | cut -f3 | sort | uniq -c` → gene, mRNA, exon, CDS, five_prime_UTR, three_prime_UTR counts



<https://phys.org/news/2026-01-ai-tool-helixer-genes-newly.html>

Why Automated Annotation Isn't Enough

- Genome annotation relies heavily on automated methods — machine learning plus rule-based/heuristic tools — but the complexity of biological systems and scarcity of reliable labeled data still trip them up.
- Genes from non-model organisms are often assigned functions by pure homology or left labeled “hypothetical gene” / “expressed protein” — offering little real biological insight.
- These inaccuracies propagate through downstream analyses, where they can be further amplified by ML/AI models trained on the flawed data.



A feedback loop: low-quality annotations degrade the reliability of both current databases and future research that depends on them.

From Manual Curation to Human–AI Teaming

Manual curation by domain experts remains essential — but it's labor-intensive and doesn't scale. AI-human teaming is the proposed way forward.

Manual Curation in Model Species

- HAVANA — Human genome
- TIGR — *Arabidopsis thaliana*
- ITAG — *Solanum lycopersicum* (tomato)

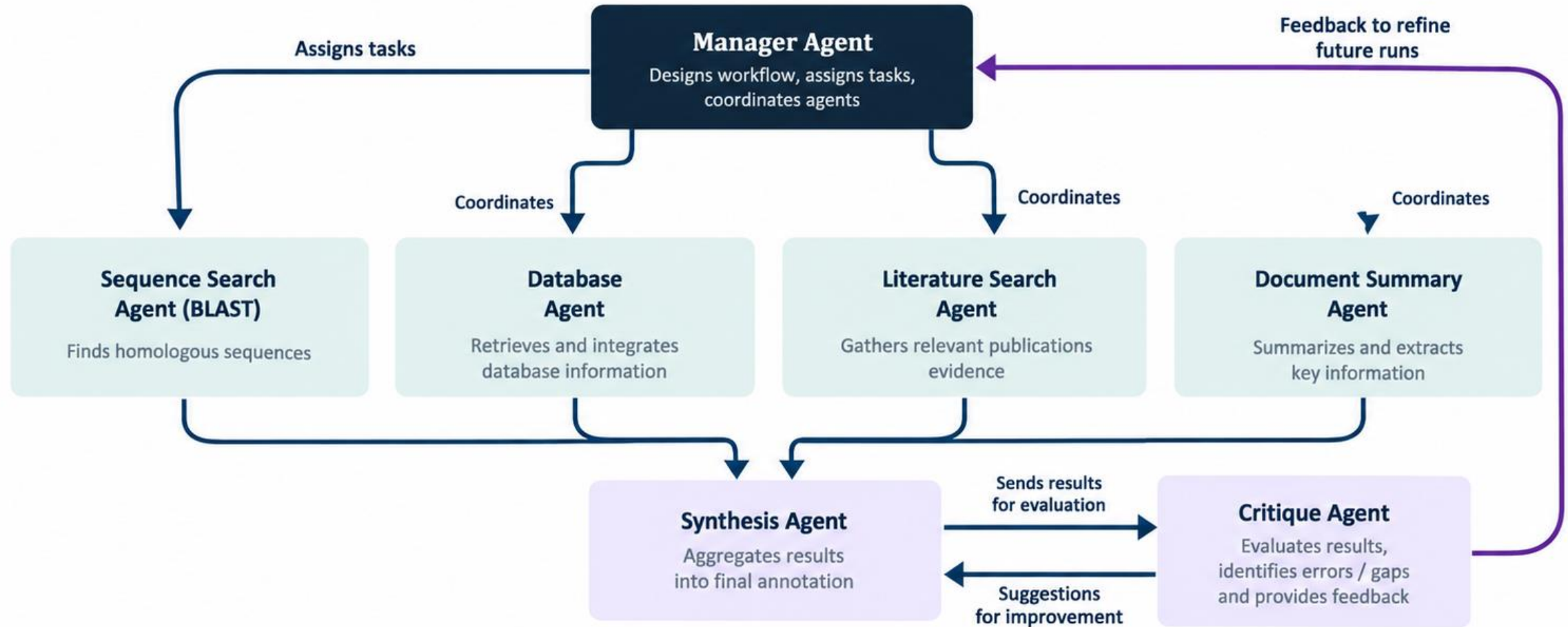
Continuously improves the accuracy and coverage of gene annotation — but requires specialized human expertise at every step.

AI's Roles in Human–AI Teaming (HAICoGA)

- **Automation**
AI independently performs the task without human intervention.
- **Augmentation**
AI enhances human experts' abilities in their own tasks.
- **Collaboration**
Humans and AI work together in coordinated effort — achieving more than either could alone.

Multi-Agent Annotation Systems

Rather than one monolithic model, HAICoGA-style systems split gene annotation across specialized AI agents — coordinated by a manager, refined by a critic.



*The manager designs the annotation workflow (e.g. BLAST → database lookup → literature → synthesis);
the critique agent's feedback loops back to refine future runs.*

The AI Agent Landscape in Genomics

From no-agent tools to small teams of two — AI assistants in genomics span a spectrum of increasing autonomy.

Tool	Agents	Task / Goal
ChatNT	No agent	Interprets DNA/RNA/protein sequences to predict gene expression, methylation, RNA stability & protein properties
DRAGON-AI	No agent	Generates ontology terms from partial user input
GeneGPT	No agent	Answers genomics questions by generating API calls to biomedical databases
Phenomics Assistant	Single agent	Natural-language querying of the Monarch phenotype knowledge graph
VarChat	Single agent	Summarizes scientific literature on specific human genomic variants
ChatGSE / biochatter	Two agents	Answers questions from knowledge graphs + literature; demoed on cell-type annotation
BioDiscoveryAgent	Two agents	Designs genetic perturbation experiments to find genes driving a phenotype
GeneAgent	Two agents	Generates — and self-verifies — biological process names for gene sets

Sources: Li et al. 2025, doi.org/10.1093/bib/bbaf377, Table 1 (AI assistants in biological and biomedical domains)

Complex Multi-Agent Systems in Genomics

With three or more agents, systems take on hierarchical team structures — a lead agent plans, specialists execute, and results are verified.

Tool	Structure	Task / Goal
BRAD	Hierarchical	Automates enrichment analysis, literature search & software pipelines
BKGAgent	Hierarchical	Checks knowledge-graph facts against literature & databases
GenoAgent	Hierarchical	Analyzes GEO/TCGA gene-expression data to find disease-associated genes
ProtAgents	Hierarchical (dynamic)	Designs novel proteins with target mechanical properties via simulation
TAIS	Hierarchical	Identifies disease-predictive genes from TCGA/GEO expression data
Virtual Lab	Hierarchical	Designs & validates nanobody binders for SARS-CoV-2 variants

All six tools above cite explainability as core to their design — from source-linked answers to fully documented, traceable multi-agent workflows.

Sources: Li et al. 2025, doi.org/10.1093/bib/bbaf377, Table 1 (AI assistants in biological and biomedical domains)

AI Architectures Powering Plant Genome Annotation

Different neural network architectures specialize in different annotation tasks — from whole-gene structure to fine-grained regulatory elements.

Tool	Task	Architecture
Helixer	Ab initio gene prediction	BiLSTM + HMM
DeepGRP	Repeat element prediction	RNN + Attention
AnnoSINE	SINE annotation	RF + DL hybrid
DNABERT	Multi-task annotation	Transformer (BERT)
iCREPCP	Cis-regulatory elements	CNN
PlantDeepSEA	Regulatory variant effects	CNN
iProm-Zea	Promoter identification	CNN

BIOINFORMATICS SOFTWARE · AMR GENOMICS

ALPAR

Automated Learning Pipeline for Antimicrobial Resistance

From raw bacterial genomes to publication-ready machine learning models — in one open-source pipeline.

Yurtseven et al., 2025 · bioRxiv 2025.07.08.663126 · Kalininalab · github.com/kalininalab/ALPAR

The AMR Challenge — and Why Computation Helps

Antimicrobial resistance (AMR) is a pressing global health threat, jeopardizing decades of progress in modern medicine. Machine learning has emerged as a powerful way to link genomic mutations to resistance phenotypes — e.g. *gyrA* SNPs and ciprofloxacin resistance is now a textbook example.

But there's a practical bottleneck: turning raw genomes into ML-ready features requires coordinating many separate tools, formats, and protocols — a real barrier for wet-lab scientists and even many bioinformaticians.



Global Threat

AMR undermines decades of antibiotic-era medical progress



WGS Is Cheap Now

Genome sequencing is routine; phenotypic screening still lags behind



Tooling Is Fragmented

Robust AMR analysis needs many coordinated bioinformatics tools



Expertise Barrier

Non-specialists struggle to assemble a correct end-to-end pipeline

Three Paradigms for AMR Prediction

Rule-Based

How it works

Predefined clinical-guideline rules (e.g. bla genes → beta-lactam resistance).

Where it struggles

Needs constant manual curation; struggles with multi-mechanism resistance (e.g. carbapenems).

GWAS

How it works

Compares resistant vs. susceptible genomes to find correlated SNPs/markers.

Where it struggles

Powerful for discovery, but limited when resistance involves complex, non-linear interactions.

Machine Learning

How it works

Learns dynamically from data; captures non-linear, multi-locus interactions.

Where it struggles

Needs careful bioinformatics pipelines — exactly the gap ALPAR is built to close.

ALPAR: A Single-Reference AMR Pipeline

ALPAR (Automated Learning Pipeline for Antimicrobial Resistance) is an **open-source Python package** that automates single-reference AMR analysis — from raw genomic sequences all the way to trained machine learning models and GWAS results.



Input

FASTA-assembled genomes + a
reference genome (GBFF)



Feature Engineering

SNPs, indels, and gene
presence/absence (GPA) — combined



Modeling

Random forest, SVM, gradient boosting
+ GWAS via Pyseer



Interface

Command-line, conda-installable, one-
command “*automatix*” run

The ALPAR Workflow, End to End



Two safeguards sit between the tables and the models:

PRPS — down-weights phylogeny-confounded mutations before training · **DataSAIL** — builds train/test splits that minimize information leakage

Also produced: phylogenetic trees (MashTree or PanACoTA/IQ-TREE) used both for PRPS and for interpreting mutation distribution.

Full documentation: github.com/kalininalab/ALPAR · alpar.readthedocs.io

The Challenges and Limitations of AI in Genomics

1

**Data Quality
& Bias**

2

**The Black-Box
Problem**

3

**Privacy &
Governance**

4

**Generalization
Gaps**

5

**The Validation
Gap**

Toward Responsible AI in Genomics

None of these challenges are reasons to avoid AI in genomics — they're the design constraints the field is actively working through.



Federated & privacy-preserving learning

Train and apply models without centralizing sensitive genomic data



Interpretable, mechanism-linked models

Move beyond black-box predictions to auditable, trustworthy reasoning



Diverse, ancestry-aware data & validation

Reduce bias by training and testing across populations, not just the best-studied ones



Sustained human oversight

Keep expert review and wet-lab validation in the loop, at every stage

AI doesn't replace genomic expertise — it scales it, provided these challenges are addressed deliberately, not as an afterthought.

“The future of biology belongs to those who embrace AI”

Programme

The same two-day programme runs at both venues (Espace Zmorda, then ISBS).

Day 1 Predict & Interpret

08:30–09:30	Registration
09:30–09:45	Welcome & Introductions
09:45–10:15	Introduction to AI for Genomics
10:15–11:00	Theoretical Introduction to AlphaFold
11:00–11:20	Break
11:20–12:30	 Practical 1: Predict
12:30–13:30	Lunch
13:30–15:00	<i>GOHREP it!</i> Interactive discussions
15:00–15:15	Break
15:15–17:00	 Practical 2: Interpret
17:00–17:15	Q&A

Day 2 Compare & Design

09:00–09:30	Day 2 Welcome & Registration
09:30–11:15	 Practical 3: Compare
11:15–11:30	Break
11:30–12:30	Model your own proteins
12:30–13:30	Lunch
13:30–14:45	 Practical 4: Design
14:45–15:05	Break
15:05–16:15	Genomics in an AI world
16:15–16:30	Summary
16:30–17:00	Q&A
17:00	Meeting close

ACKNOWLEDGEMENTS

The Sainsbury Laboratory
TSL



Thank You!



- Amal Boukteb
- James Canham
- Sophien Kamoun
- Andy Posbeyikian
- Yu Sugihara
- Our facilitators!

